EE618

Assignment-2, Part-II

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1 Motivation behind the topology

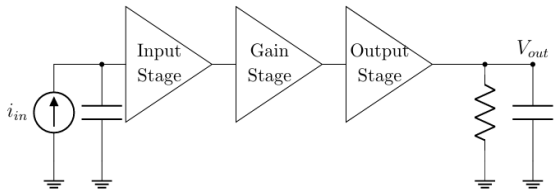


Figure 1: High-Level Diagram of the TIA

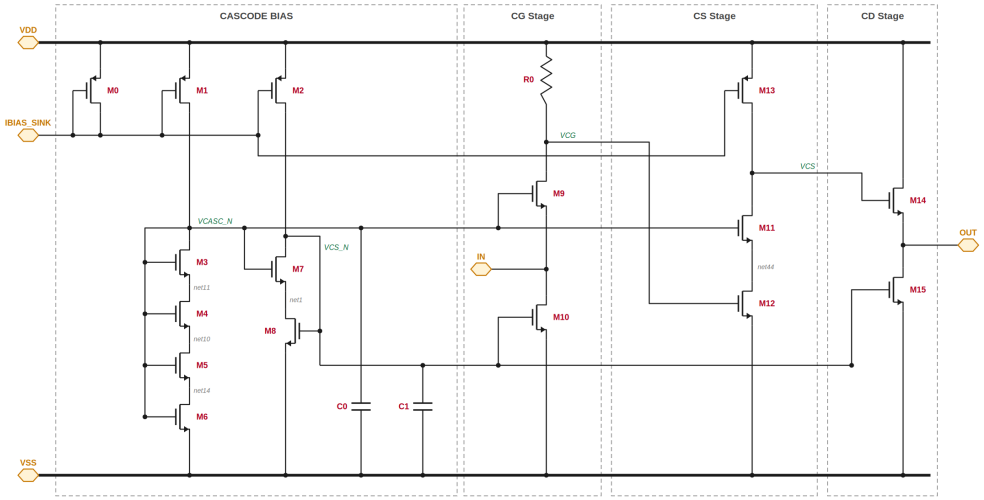


Figure 2: Schematic of the TIA

I used a three-stage cascade of a common-gate (CG) input stage, a common-source (CS) gain stage, and a common-drain (CD) output stage (Design Suggestion 1, figure above). All stages are biased from a single cascode-bias block fed by the one ideal current source at the `IBIAS` pin. The reasons for this choice are given below.

1. The design needs $|Z_O| > 31.6\,\mathrm{k\Omega}$ (90 dB) and $BW > 70\,\mathrm{MHz}$, driving a $10\,\mathrm{k\Omega} \parallel 1\,\mathrm{pF}$ load from a source with $C_{in} = 500\,\mathrm{fF}$. A single stage cannot do all of this. Splitting the design into a current-sensing stage, a gain stage and a buffer lets each stage be optimised for its own role (low R_{in}, high gain, low R_{out}). The overall transimpedance is $Z_O \approx R_D \cdot A_{v,CS} \cdot A_{v,CD}$, where R_D is the CG load resistor, $A_{v,CS}$ is the CS gain and $A_{v,CD} \lesssim 1$.

2. **CG input stage (current sensing and bandwidth).** *Purpose:* The input pole is set by C_{in} and the resistance seen at the input node. A CG device presents $R_{in} \approx 1/g_m$ (low), so $f_{p,in} \approx g_m/(2\pi C_{in})$ is pushed to high frequency. The CG stage also converts i_{in} to a voltage with a well-defined gain. *Implementation:* The CG nMOS has its source at `IN` and its gate tied to the cascode-bias voltage `V_CASC_N`. Below it, a second nMOS (gate at `V_CS_N`) sinks the DC current, so no DC component is needed in i_{in}. The drain load is a poly resistor R_D to V_{DD}, which gives $V_{CG} \approx -i_{in}R_D$.

3. **Resistor load in the CG stage.** A resistor gives a linear, PVT-tolerant transimpedance, contributes less distortion than an active load, and costs little headroom at $V_{DD} = 1.1\,\mathrm{V}$. Its drawback is the DC drop $I_D R_D$. This limits R_D and is the reason the remaining gain comes from the next stage.

4. **CS gain stage (high gain).** *Purpose:* Provide the bulk of the 90 dB gain. *Implementation:* An nMOS CS device is driven by V_{CG} and loaded by a pMOS current source (gate from the pMOS mirror bias) and the nMOS bias device below it (gate at `V_CS_N`). The gain is $A_{v,CS} \approx -g_{m,n}(r_{o,n} \parallel r_{o,p})$. An active current-source load gives a large small-signal load resistance in little area, and the long-channel devices raise r_o further.

5. **CD (source-follower) output stage.** *Purpose:* The CS output node is high impedance and cannot drive $10\,\mathrm{k\Omega} \parallel 1\,\mathrm{pF}$ directly without losing gain and bandwidth. A buffer isolates it. *Implementation:* An nMOS source follower (gate from the CS output V_{CS}) with an nMOS current-sink (gate at `V_CS_N`) sets $R_{out} \approx 1/g_m$ and a gain close to unity. The output pole is $f_{p,out} \approx g_m/(2\pi C_L)$ to meet the 70 MHz target.

6. **Single shared bias network (cascode bias block).** Only one ideal current source (`IBIAS`) is permitted, so a pMOS mirror copies it into diode-connected nMOS stacks that generate `V_CASC_N` and `V_CS_N`. These two voltages bias every stage, so the stage currents track each other over process and temperature, and the bias cost is a mirror ratio. The decoupling capacitors on these nodes keep them quiet and stable at high frequency. The mirror ratios set the branch currents I_{CG}, I_{CS} and I_{CD}, which appear in the total power and is the quantity to minimise in the FOM.

7. **DC-coupled chain and the bias stabiliser.** The stages are directly coupled ($V_{CG} \to$ CS gate, $V_{CS} \to$ CD gate), with no coupling capacitors, so the full-bandwidth signal path is not broken and the input node is biased without a DC input current. The price is that the high-gain, high-impedance nodes make the DC operating point extremely sensitive to mismatch and drift. The ideal *Bias Stabiliser* in the testbench fixes this. It low-pass filters the DC output through $R = 1\,\mathrm{M\Omega}$ and $C = 1\,\mathrm{F}$ (corner $\approx 0.16\,\mu\mathrm{Hz}$), compares it with V_{REF} (the target output DC level), and injects a correcting current $I = g_m(V_{OUT,DC} - V_{REF})$, back into the `IN` node. This is a negative-feedback loop whose sign must be correct, otherwise the bias saturates at a rail. Its corner is far below the 1 MHz measurement frequency, so it sets the DC point without affecting $|Z_O|$, the bandwidth or the peaking.

2 Hand Calculations

2.1 Initial sizing

The transimpedance of the cascade is

$$Z_O \approx R_D \, A_{v,CS} \, A_{v,CD},$$

where R_D is the CG load resistor (the CG stage converts i_{in} to $V_{CG} \approx -i_{in} R_D$), $A_{v,CS}$ is the gain of the CS stage and $A_{v,CD} \lesssim 1$ is the follower gain. As a first pass I took $A_{v,CD} \approx 1$ and approximated the CS gain by the intrinsic gain of the nMOS alone, i.e. I neglected the pMOS load:

$$Z_O \approx g_m \, r_o \, R_D, \qquad g_m = \frac{2I_D}{V_{OV}}, \qquad r_o = \frac{1}{\lambda I_D} \quad \Rightarrow \quad Z_O \approx \frac{2}{\lambda V_{OV}} \, R_D.$$

Note that the intrinsic gain $g_m r_o = 2/(\lambda V_{OV})$ is independent of I_D. I started with $L = 200\,\text{nm}$ and $V_{OV} = 0.2\,\text{V}$ for all transistors. From the process table,

$$\lambda_n = 0.12 + \frac{0.065}{0.2} = 0.445\,\text{V}^{-1}, \qquad \lambda_p = 0.06 + \frac{0.04}{0.2} = 0.26\,\text{V}^{-1},$$

$$(g_m r_o)_n = \frac{2}{0.445 \times 0.2} = 22.5, \qquad (g_m r_o)_p = \frac{2}{0.26 \times 0.2} = 38.5.$$

Targeting $Z_O = 40\,\text{k}\Omega$ (against the $31.6\,\text{k}\Omega$ spec, to leave margin):

$$R_D = \frac{Z_O}{(g_m r_o)_n} = \frac{40\,\text{k}}{22.5} \approx 1.78\,\text{k}\Omega.$$

The CS device M12 is gated by V_{CG}, so for it to have $V_{OV} = 0.2\,\text{V}$ we need

$$V_{CG} = V_{TH} + V_{OV} = 0.33 + 0.2 = 0.53\,\text{V}.$$

Since $V_{CG} = V_{DD} - I_D R_D$, the drop across the load resistor is fixed:

$$I_D = \frac{V_{DD} - V_{CG}}{R_D} = \frac{1.1 - 0.53}{1.78\,\text{k}\Omega} \approx 0.32\,\text{mA}.$$

The same $I_D R_D \approx 0.57\,\text{V}$ budget was kept when R_D was later raised $5\times$ (the CG-branch current was cut $5\times$ to $64\,\mu\text{A}$).

This gave $I_D = 0.32\,\text{mA}$. Using the square law, with $\mu_n C_{ox} = 480\,\mu\text{A/V}^2$,

$$W_n = \frac{2 I_D L}{\mu_n C_{ox} V_{OV}^2} = \frac{2(0.32\,\text{m})(0.2\,\mu)}{(480\,\mu)(0.2)^2} \approx 6.6\,\mu\text{m}$$

Similarly,

$$W_p = 10.66\,\mu\text{m}.$$

These widths were used for all nMOS and pMOS devices, respectively.

2.2 Observation and re-tuning

Simulation gave much less gain than this first-pass estimate. Probing the gain of each stage separately showed that the CS stage gave sufficient gain, but the CG stage gave much less.

This is explained by the first-pass model. The CS stage is loaded by the pMOS current source, so its gain is $A_{v,CS} = g_m(r_{o,n} \parallel r_{o,p})$, not $(g_m r_o)_n$:

$$A_{v,CS} = \frac{2}{(\lambda_n + \lambda_p)V_{OV}} = \frac{2}{0.705 \times 0.2} \approx 14.2 \; < \; (g_m r_o)_n = 22.5.$$

With $R_D = 1.78\,\mathrm{k\Omega}$ the corrected estimate is

$$Z_O \approx R_D\, A_{v,CS} = 1.78\,\mathrm{k} \times 14.2 \approx 25\,\mathrm{k\Omega}\ (\approx 88\,\mathrm{dB}),$$

which is below the 90 dB spec even before the follower loss. The first-pass design was therefore short of the spec, and the CG load resistor R_D was the limiting term. The changes below were made in response.

1. **CG stage (gain).** The CG stage transimpedance scales with R_D, so R_D was increased 5× to 8.9 kΩ. To keep the DC drop $I_D R_D \approx 0.57\,\mathrm{V}$ unchanged (and so keep the devices in saturation), the current in this branch was reduced 5× by scaling the widths of the two CG-stage nMOS down by 5×: $W = 6.6/5 = 1.32\,\mu\mathrm{m}$.

   $$I_{CG} = \frac{0.32\,\mathrm{mA}}{5} = 64\,\mu\mathrm{A}, \qquad I_{CG}R_D = 64\,\mu \times 8.9\,\mathrm{k} \approx 0.57\,\mathrm{V}.$$

   The overdrive is unchanged, so $A_{v,CS}$ is unchanged and the transimpedance rises 5×, from 25 kΩ to

   $$Z_O \approx R_D\, g_m(r_{o,n} \parallel r_{o,p}) = 8.9\,\mathrm{k} \times \frac{2}{0.705 \times 0.2} \approx 126\,\mathrm{k\Omega}\ (\approx 102\,\mathrm{dB}),$$

   before the loss in the CD stage ($A_{v,CD} < 1$ because of R_L and the follower's finite g_m and body effect) and other second-order loading. The simulated value of about 99 dB is within 3 dB of this estimate, and above the 90 dB spec.

2. **CD stage (headroom).** The lower nMOS of the CD stage was entering the linear region because its V_{DS} headroom was too small. Its width was scaled down 3× ($W = 2.2\,\mu\mathrm{m}$), which reduces the branch current and the follower's V_{GS}, and raises the output DC level, restoring V_{DS} headroom.

3. **Power optimisation.** With $I_{BIAS,SINK} = I_D$ everything was biased correctly, but to save power the reference current was reduced 3× to $I_{BIAS,SINK} \approx 110\,\mu\mathrm{A}$. The pMOS mirror device carrying this current was scaled by the same factor, $W_p = 10.66/3 \approx 3.55\,\mu\mathrm{m}$, so that its overdrive is unchanged.

4. **Capacitors.** The two bias-decoupling capacitors were set in the pF range, so that their poles, $f = 1/(2\pi RC)$ with R the (large) resistance of the bias node, fall well below the frequencies of interest and do not affect the response in the passband.

Bandwidth estimate: The bandwidth is set by the two nodes that carry large capacitance, the input node ($C_{in} = 500\,\text{fF}$) and the output node ($C_L = 1\,\text{pF}$). The internal nodes V_{CG} and V_{CS} carry only about 20 fF each, so their poles lie well above 500 MHz.

Input pole. With $I_{CG} = 64\,\mu\text{A}$ and $V_{OV} = 0.2\,\text{V}$, $g_{m,9} = 2I_{CG}/V_{OV} = 640\,\mu\text{S}$, so $R_{in} \approx 1/g_{m,9} \approx 1.56\,\text{k}\Omega$:

$$f_{p,in} \approx \frac{g_{m,9}}{2\pi C_{in}} = \frac{640\,\mu}{2\pi(500\,\text{f})} \approx 200\,\text{MHz}.$$

Output pole. The follower carries about 107 µA, so $g_{m,14} \approx 1.07\,\text{mS}$ and the output resistance is $(1/g_{m,14}) \parallel R_L \approx 855\,\Omega$:

$$f_{p,out} \approx \frac{1}{2\pi(855)(1\,\text{pF})} \approx 185\,\text{MHz}.$$

Internal poles. $f_{p,CG} \approx \dfrac{1}{2\pi R_D C_{CG}} \approx 745\,\text{MHz}$ with $C_{CG} \approx 24\,\text{fF}$ (mainly C_{gs12}), and $f_{p,CS} \approx 690\,\text{MHz}$ with $R \approx 11\,\text{k}\Omega$ and $C \approx 21\,\text{fF}$.

Combining the four poles,

$$f_{-3\text{dB}} \approx \left[\frac{1}{200^2} + \frac{1}{185^2} + \frac{1}{745^2} + \frac{1}{690^2}\right]^{-1/2} \approx 130\,\text{MHz},$$

which is above the 70 MHz specification and close to the simulated 125 MHz.

3 Final Component Parameters

3.1 Summary of all Sizes

Table 1: Sizes of all components (as labelled in the schematic below) in the TIA core

Block	Device	Type	W (µm)	L (nm)	Role / Remarks
Bias	M0	pch	3.55	200	diode-connected, carries $I_{BIAS_SINK} \approx 110\,\mu A$
	M1	pch	10.66	200	mirror (3× M0), feeds M3–M6 stack, generates `VCASC_N`
	M2	pch	10.66	200	mirror (3× M0), feeds M7/M8 branch, generates `VCS_N`
	M3–M6	nch	6.6	200	series-stacked, common-gate ladder setting `VCASC_N`
	M7	nch	6.6	200	cascode device, gate at `VCASC_N`
	M8	nch	6.6	200	diode-connected reference for `VCS_N`
	C0, C1	nmoscap_25	25	25 µm	value = 2.52 pF
CG stage	M9	nch	1.32	200	CG device, gate at `VCASC_N`, source at `IN`
	M10	nch	1.32	200	DC sink at `IN`, gate at `VCS_N` ($\approx 64\,\mu A$)
	R0	rppolywo	2	22.85 µm	value = 8.9 kΩ
CS stage	M12	nch	6.66	200	CS device, gate at `VCG`
	M11	nch	6.6	200	cascode device, gate at `VCASC_N`
	M13	pch	10.66	200	current-source load, gate from the pMOS mirror
CD stage	M14	nch	6.6	200	source follower, gate at `VCS`
	M15	nch	2.2	200	current sink, gate at `VCS_N` (reduced 3×)

3.2 Screenshots with Component Parameters Annotated

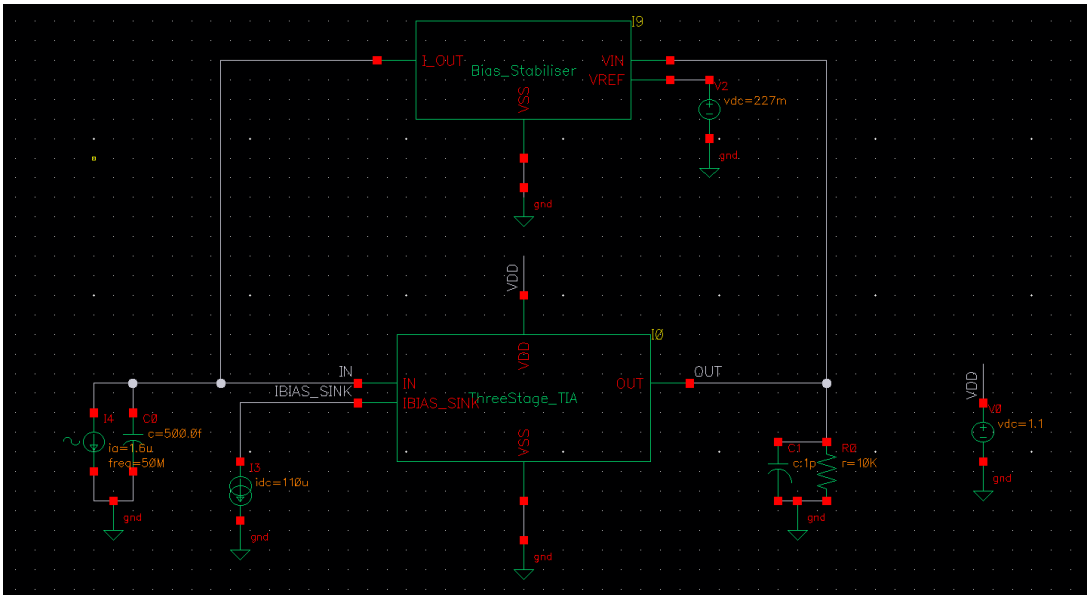


Figure 3: Component Parameters in the TIA Testbench

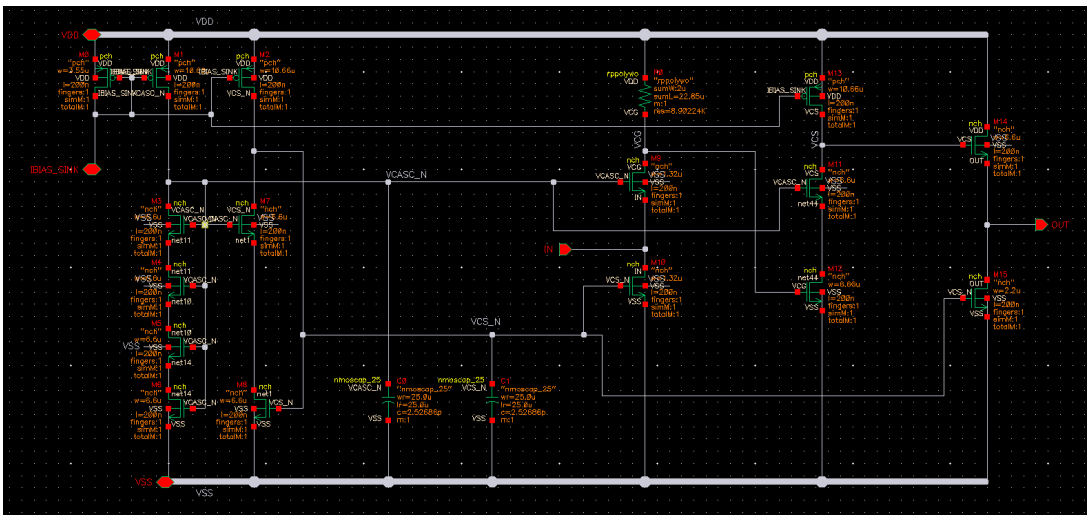


Figure 4: Component Parameters in the TIA Design

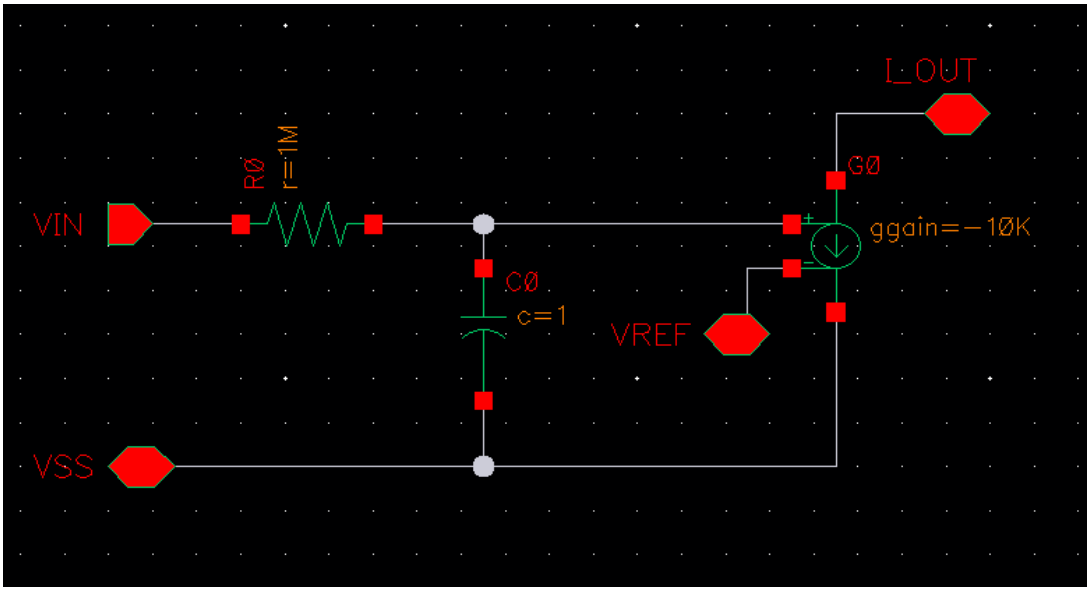


Figure 5: Component Parameters in the Bias Stabiliser

4 Results at 0^oC

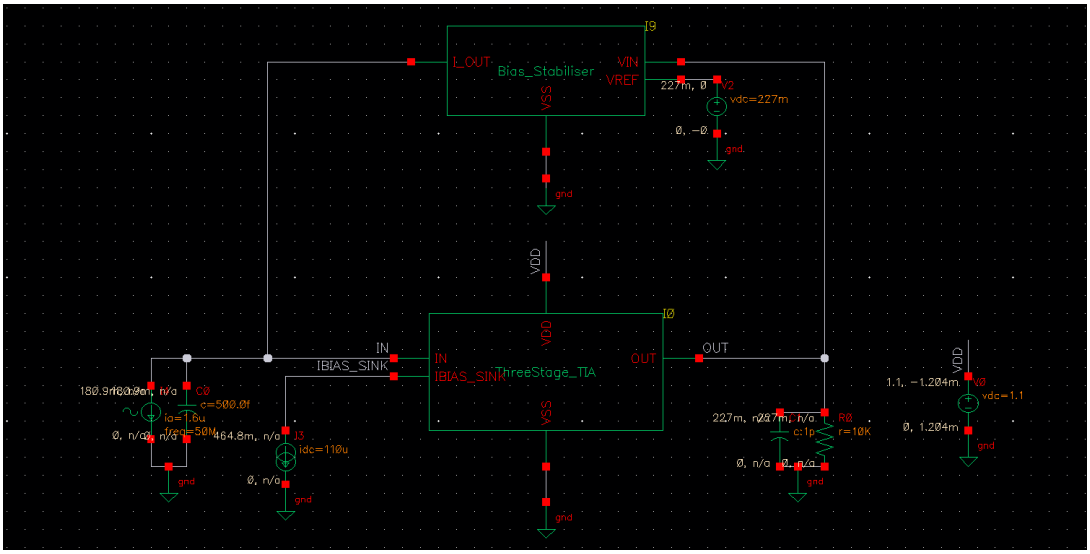


Figure 6: DC Voltages and Currents Annotated in the TIA Testbench

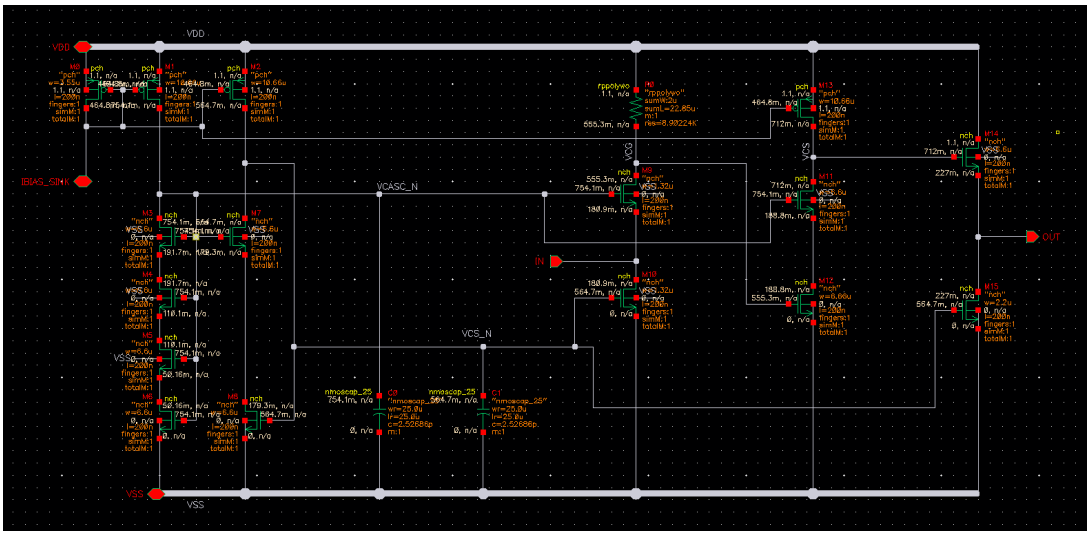


Figure 7: DC Voltages and Currents Annotated in the TIA Design

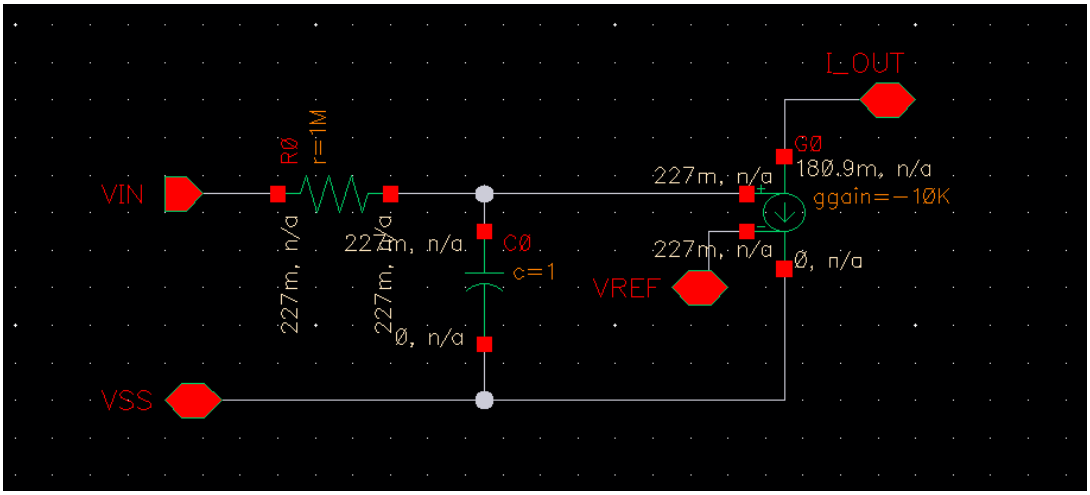


Figure 8: DC Voltages and Currents Annotated in the Bias Stabiliser

Outputs Setup | Run Preview

Name	Type	Details	Value	Plot	Plot Target	Save	Spec
dB20	expr	dB20(VF("/OUT"))					
BW	expr	bandwidth(VF("/OUT") 3 "low")	127.7M				
dB20_max	expr	ymax(dB20(VF("/OUT")))	99.82				
dB20_CG	expr	dB20(VF("/I0/VCG"))					
dB20_CS	expr	dB20(VF("/I0/VCS"))					
Power	expr	(IDC("/I0/VDD") * 1.1)	1.324m				
Input_trans	expr	VT("/IN")					
Output_trans	expr	VT("/OUT")					
Output_trans_pp	expr	peakToPeak(VT("/OUT"))	251m				
Peaking	expr	(ymax(dB20(VF("/OUT"))) - value(dB20(VF("/OUT")) 1000000.0))	250.7u				

Figure 9: Results

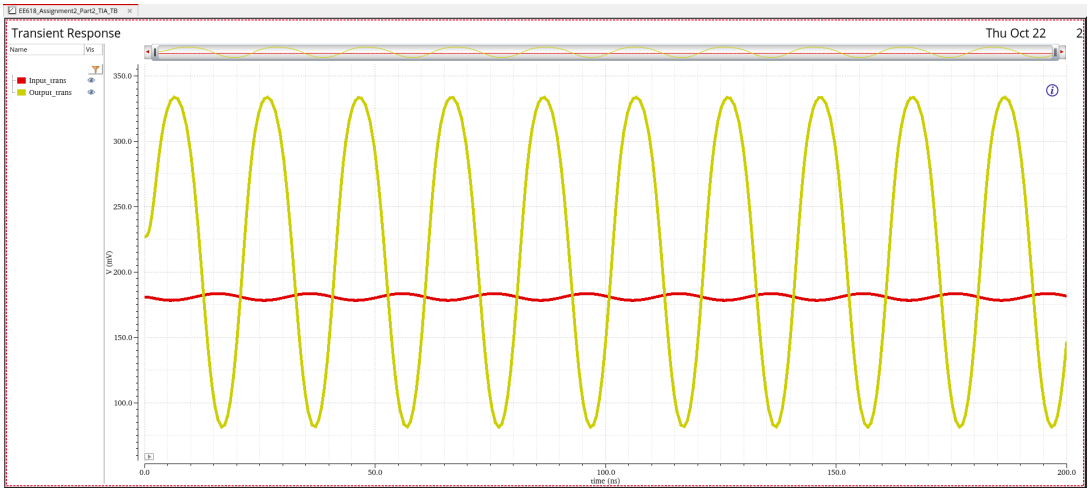


Figure 10: Transient Response (Input and Output Voltages)

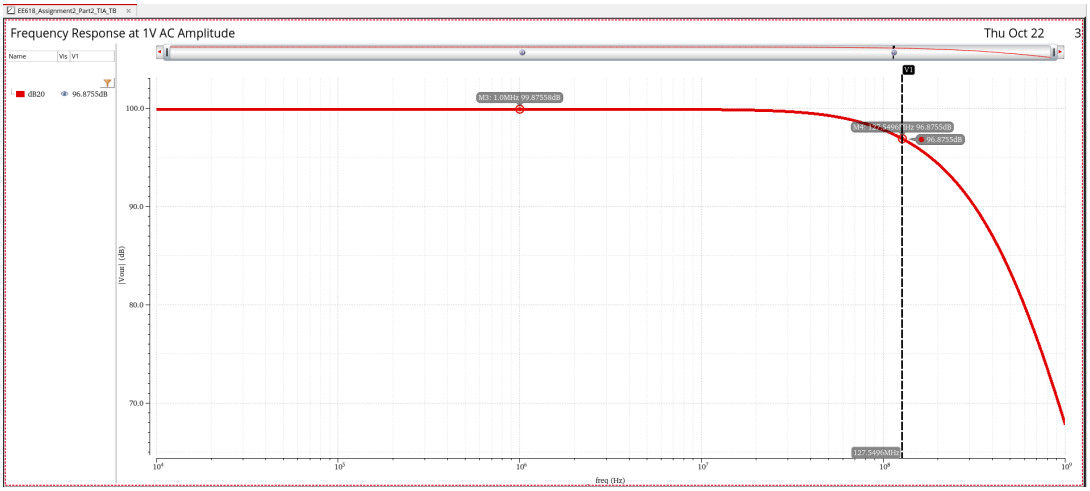


Figure 11: Frequency Response at 1V AC Amplitude with $|V_{out}|$ in $db\Omega$ at 1MHz and -3dB Magnitude Frequency annotated

5 Results at 27^oC

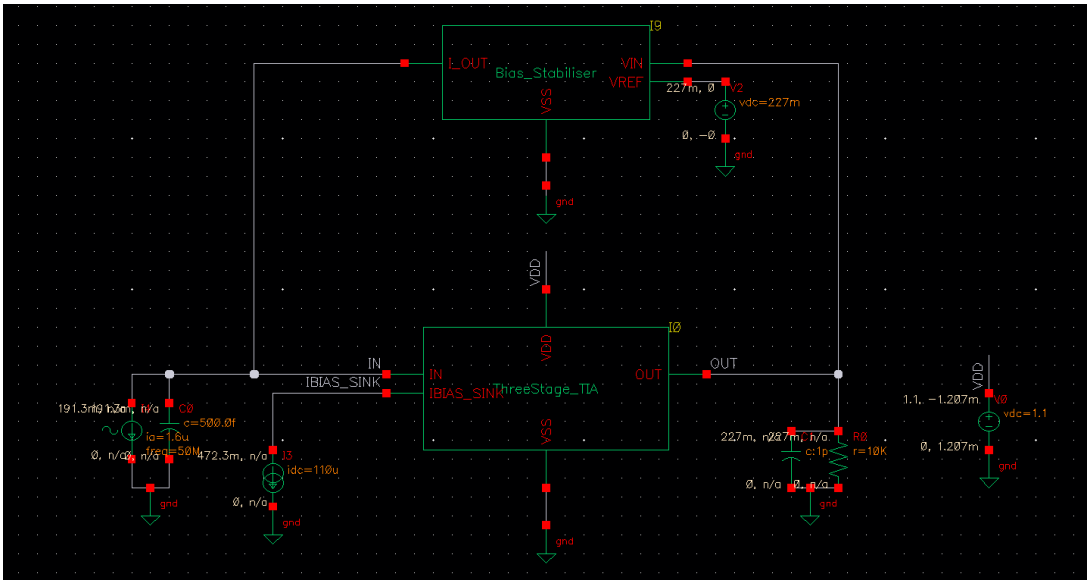


Figure 12: DC Voltages and Currents Annotated in the TIA Testbench

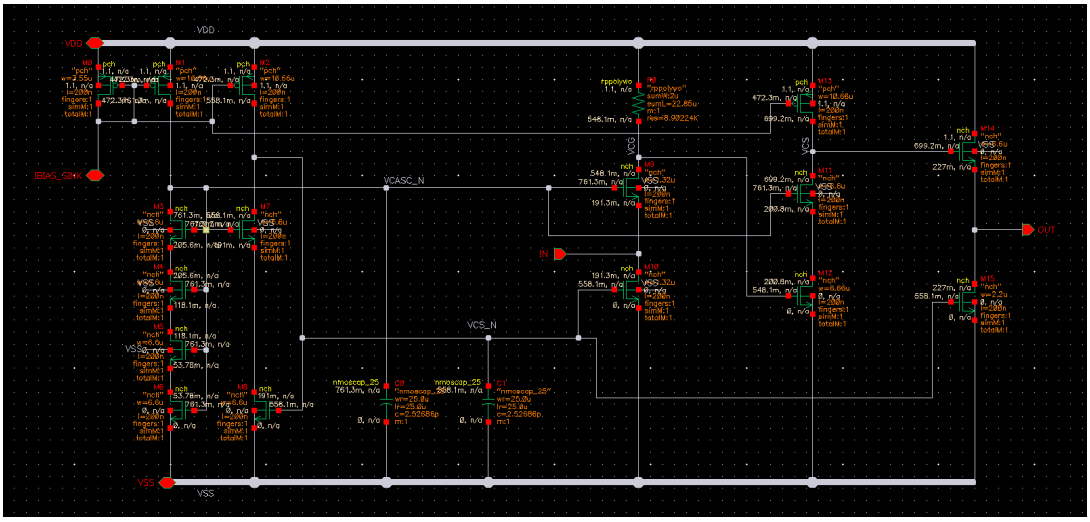


Figure 13: DC Voltages and Currents Annotated in the TIA Design

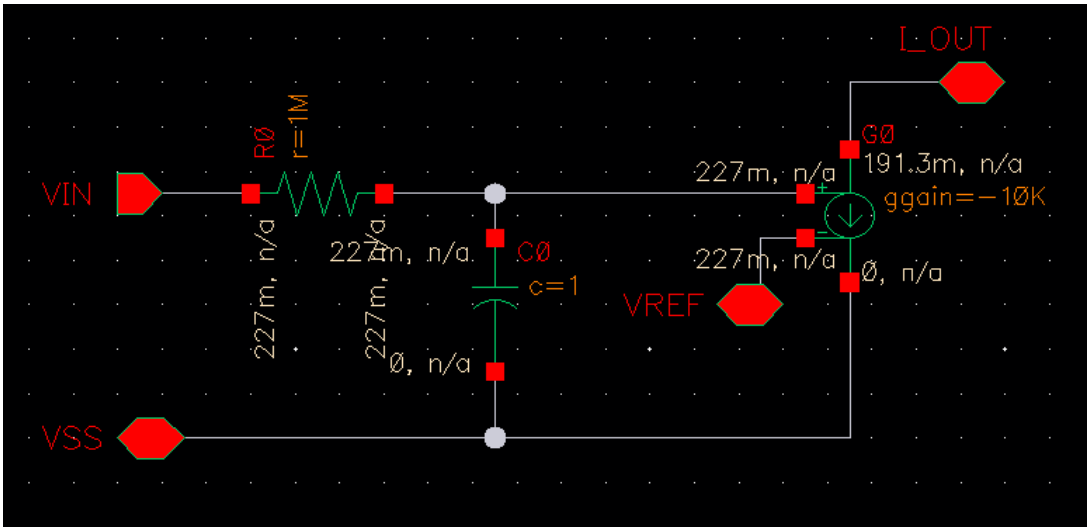
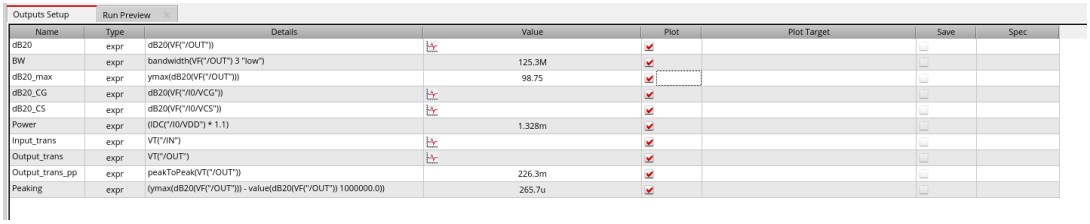


Figure 14: DC Voltages and Currents Annotated in the Bias Stabiliser



Outputs Setup | Run Preview

Name	Type	Details	Value	Plot	Plot Target	Save	Spec
dB20	expr	dB20(VF("/OUT"))					
BW	expr	bandwidth(VF("/OUT") 3 "low")	125.3M				
dB20_max	expr	ymax(dB20(VF("/OUT")))	98.75				
dB20_CG	expr	dB20(VF("/I0/VCG"))					
dB20_CS	expr	dB20(VF("/I0/VCS"))					
Power	expr	(IDC("/I0/VDD") * 1.1)	1.328m				
Input_trans	expr	VT("/IN")					
Output_trans	expr	VT("/OUT")					
Output_trans_pp	expr	peakToPeak(VT("/OUT"))	226.3m				
Peaking	expr	(ymax(dB20(VF("/OUT"))) - value(dB20(VF("/OUT")) 1000000.0))	265.7u				

Figure 15: Results

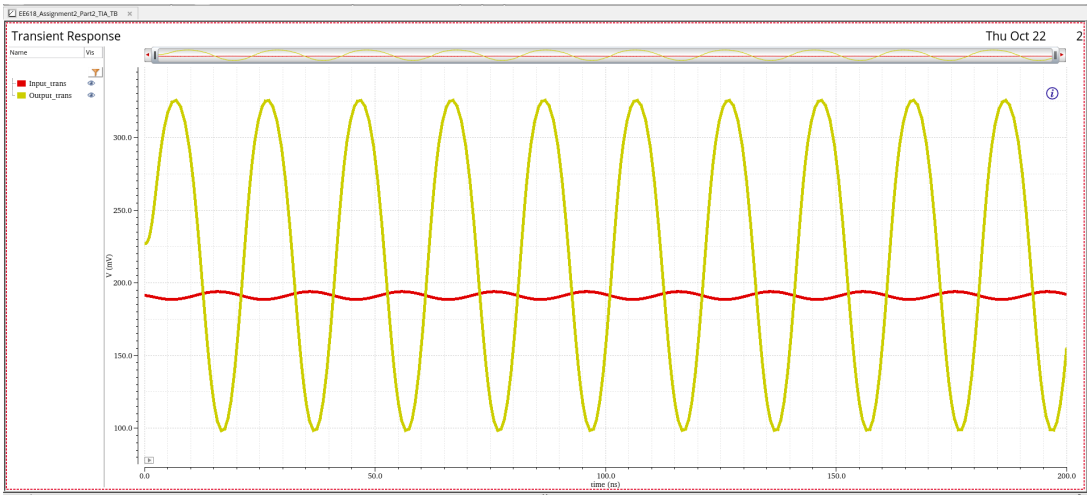


Figure 16: Transient Response (Input and Output Voltages)

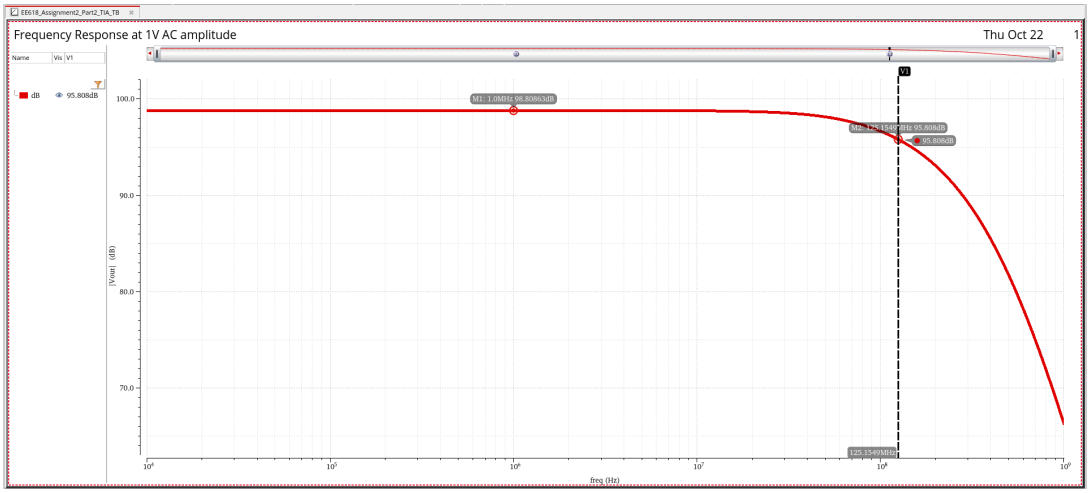


Figure 17: Frequency Response at 1V AC Amplitude with $|V_{out}|$ in $db\Omega$ at 1MHz and -3dB Magnitude Frequency annotated

6 Results at 100^oC

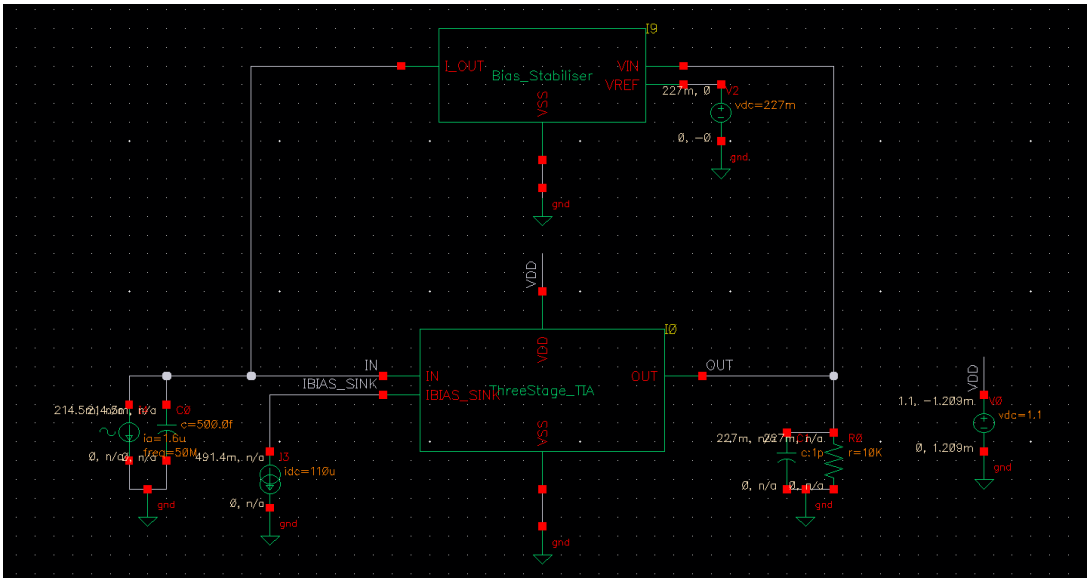


Figure 18: DC Voltages and Currents Annotated in the TIA Testbench

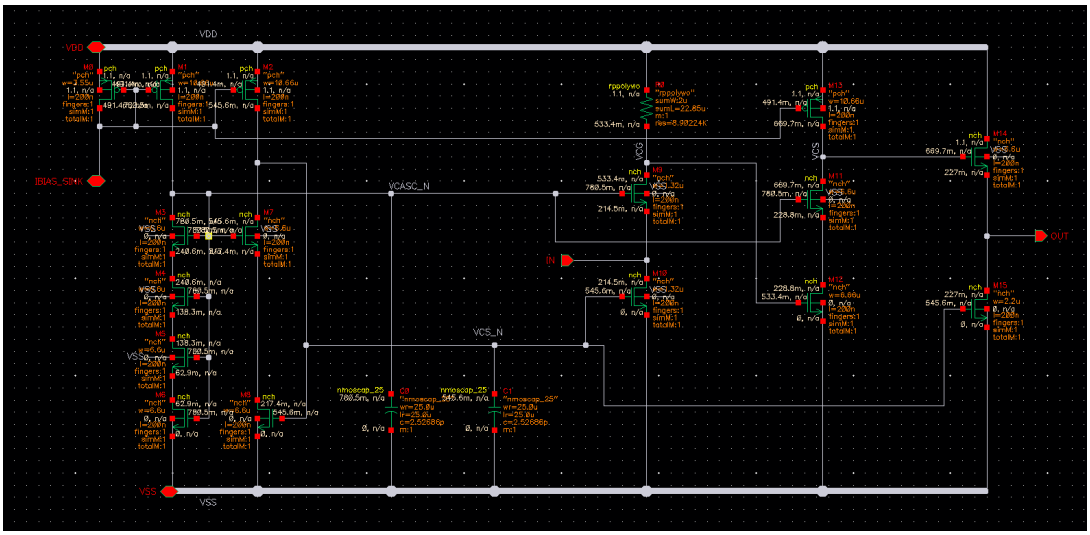


Figure 19: DC Voltages and Currents Annotated in the TIA Design

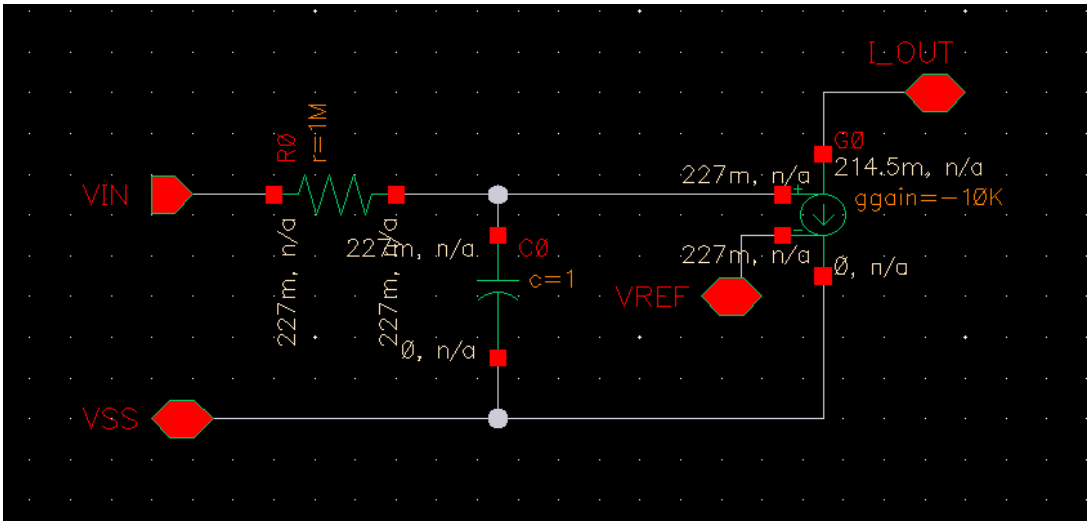
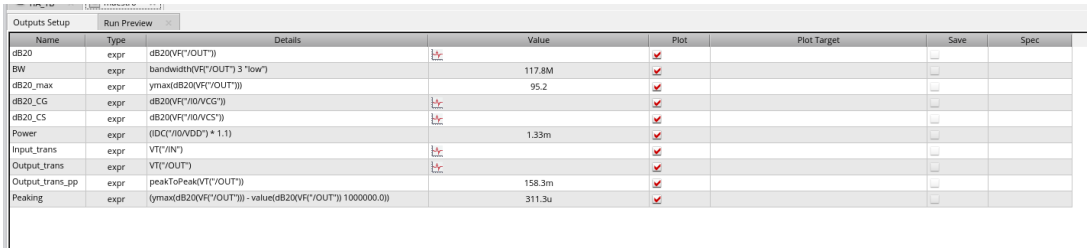


Figure 20: DC Voltages and Currents Annotated in the Bias Stabiliser



Outputs Setup | Run Preview

Name	Type	Details	Value	Plot	Plot Target	Save	Spec
dB20	expr	dB20(VF("/OUT"))		☑			
BW	expr	bandwidth(VF("/OUT") 3 "low")	117.8M	☑			
dB20_max	expr	ymax(dB20(VF("/OUT")))	95.2	☑			
dB20_CG	expr	dB20(VF("/I0/VCG"))		☑			
dB20_CS	expr	dB20(VF("/I0/VCS"))		☑			
Power	expr	(IDC("/I0/VDD") * 1.1)	1.33m	☑			
Input_trans	expr	VT("/IN")		☑			
Output_trans	expr	VT("/OUT")		☑			
Output_trans_pp	expr	peakToPeak(VT("/OUT"))	158.3m	☑			
Peaking	expr	(ymax(dB20(VF("/OUT"))) - value(dB20(VF("/OUT")) 1000000.0))	311.3u	☑			

Figure 21: Results

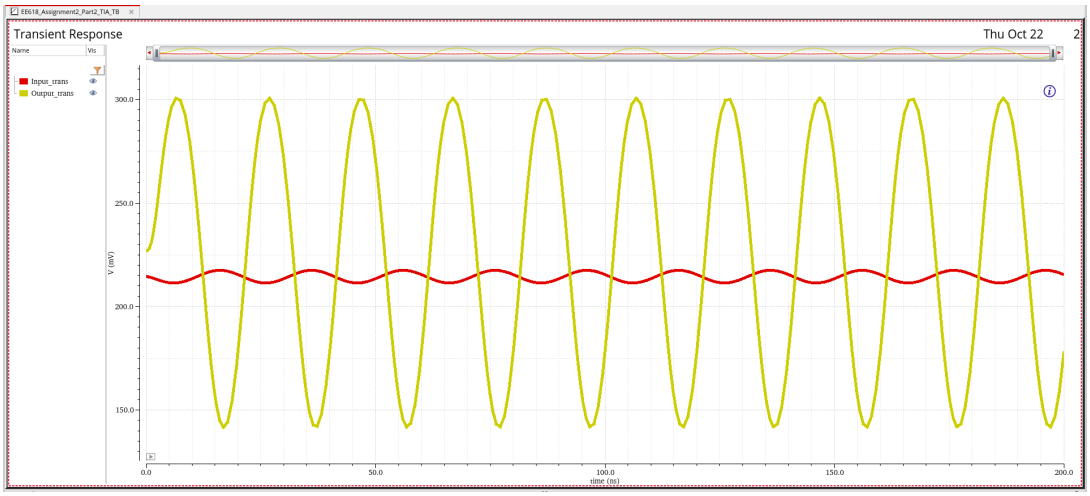


Figure 22: Transient Response (Input and Output Voltages)

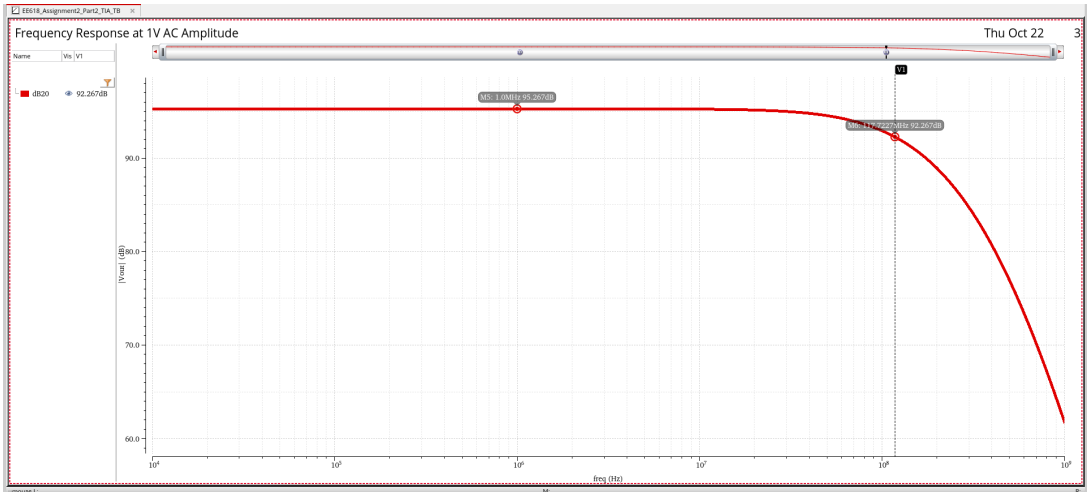


Figure 23: Frequency Response at 1V AC Amplitude with $|V_{out}|$ in $db\Omega$ at 1MHz and -3dB Magnitude Frequency annotated

7 Summary of Results at various temperatures

Table 2: Performance metrics at different temperatures

Parameter	**0°C**	**27°C**	**100°C**
Bandwidth (MHz)	127.6	125.2	117.7
$dB_{20,\max}$ (dB)	99.88	98.81	95.27
Peaking (dB)	250.7μ	265.7μ	311.3μ
$V_{\text{out,pp}}$ (mV)	252.3	227.3	159.1
Power (mW)	1.324	1.328	1.330

$$
\begin{aligned}
A_{\text{est}}\,(\mu\text{m}^2) &= \sum_{\text{transistors}} W(L + 0.35\,\mu\text{m}) + \sum_{\text{resistors}} \text{segments}(W \times L) + \sum_{\text{capacitors}} \text{multiplier}(W \times L) \\
&= 54.8735 + 45.7 + 1250 \\
&= \boxed{1350.5735\ \mu\text{m}^2}
\end{aligned}
$$

$$
\begin{aligned}
\text{FOM at } 27^\circ\text{C} &= \frac{\text{Yield(Worst-Case)}}{A_{\text{est}}\,(\mu\text{m}^2) \times \text{PWR}\,(\mu\text{W})} \\
&= \frac{1}{1350.5735 \times 1328} \\
&= \boxed{5.575 \times 10^{-7} \mu\text{m}^{-2}\mu\text{W}^{-1}}
\end{aligned}
$$

8 Results of Global Monte-Carlo Simulation at 27^oC

Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 100 %(156 passed/156 pts) Confidence Level: <not set> Filter: <not set>									
EE618_Assignment2_Part2_TIA_TB_Global_MC									
	BW	100% (156/156)	113M	> 70M	142.1M	125.5M	5.563M	3.32	0
	dB20_max	100% (156/156)	96.62	> 90	100.7	98.75	790.2m	3.69	0
	Output_trans_pp	100% (156/156)	184.1m	> 100m	271.5m	226.9m	16.5m	2.56	0
	Peaking	100% (156/156)	210.7u	< 1	329.5u	266.7u	23.13u	1.44e+04	0

Figure 24: Results of Global MC Simulation showing all yields

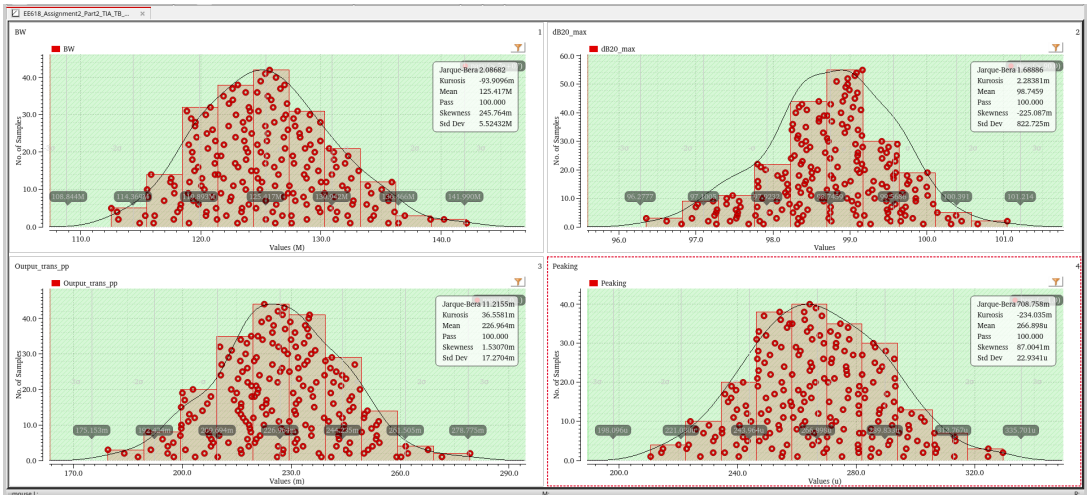


Figure 25: Histograms of BW, Max Gain, Output peak-to-peak, Peaking

9 Challenges Faced

My implementation **met all the requirements**. The following difficulties were faced during the implementation:

1. **Too many knobs to tune:** With so many transistors, resistors and bias branches, it was hard to tell which W, L or bias-current change was responsible for a given shift in gain, bandwidth or DC point, so I had to tune one parameter at a time and log its effect.
2. **Transistors in CS Amplifier dropping out of saturation:** The stacked, directly coupled stages leave little headroom at $V_{DD} = 1.1\,\mathrm{V}$, and a small change in one node voltage pushed neighbouring transistors into triode, collapsing the gain.
3. **Huge bandwidth but low gain:** Early iterations gave bandwidth well above 70 MHz but a $|Z_O|$ far below 90 dB, since the low-impedance nodes that raise bandwidth also lower gain.
4. **Bias point sensitivity:** The high-impedance DC-coupled nodes drifted easily with small mismatch or temperature changes, so the Bias Stabiliser needed to hold V_{OUT} at V_{REF} without saturating at a rail.